TalksTry — Calling System Bug Report & Fix Guide

Prepared: 26 October 2026
Analysis window: 12–26 October 2026 (14 days)
Data sources: mediasoup call-server pm2 logs (57,491 lines) · `talsksty-baackend/logs/calls/` (7,633 lines) · MongoDB `calllogs` + `calltraces` (649 outgoing calls)

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1. Executive summary

Do bugs mile. Dono ka root cause confirm ho gaya hai with hard data.

#	Bug	Impact	Root cause	Fix location
1	**One-way audio** — caller ki awaaz callee ko nahi jaati, callee ki caller ko jaati hai	**52 of 509 accepted calls (~10.2%)**	Caller ka mic server pe `paused` reh jaata hai. Use un-pause karne wala **sirf ek** socket event hai (`callAccepted`). Woh miss hua to kuch bhi self-heal nahi karta	Mobile CallingSDK + `talkstry-web`
2	**Call apne aap cancel ho jaati hai**	**232 of 649 outgoing calls (35.7%)**	APNs VoIP push ka pehla attempt **72% baar ~2.08s** me fail hota hai (IPv6 happy-eyeballs toot raha hai). Callee ka phone ~2.3s pe bajta hai, aur caller ki app ~2.3s pe dial abort kar deti hai	**`talsksty-baackend` — 1 line**

Platform traffic (kahan asar hoga)

Platform	Completed	Cancelled	Cancel rate
Android	204	140	**39%**
iOS	137	88	**34%**

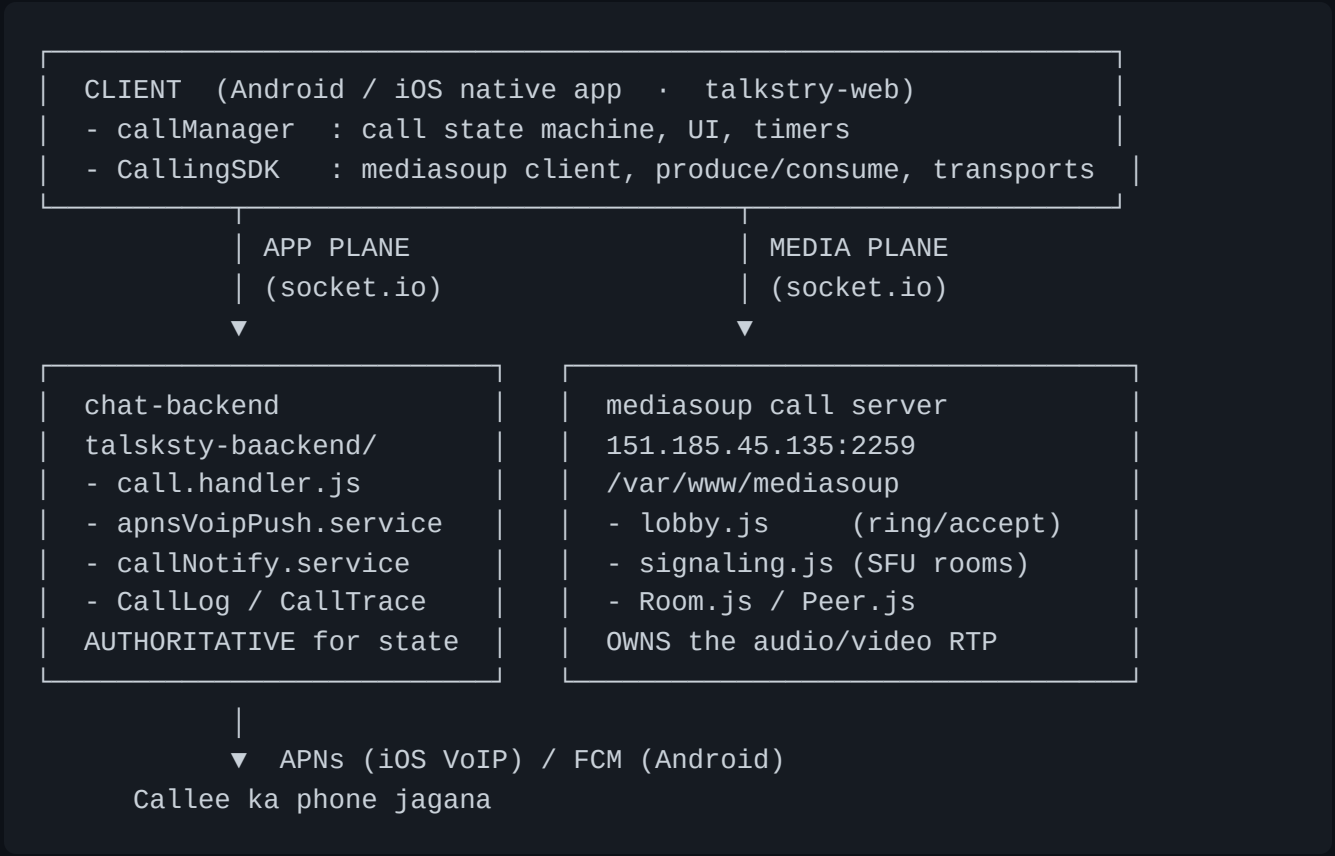
Platform	Completed	Cancelled	Cancel rate
Web	6	4	—

> Production traffic **99% native app** hai. Web pe sirf 10 calls. Isliye asli impact backend + mobile SDK fix se aayega, `talkstry-web` se nahi.

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2. Architecture — teen planes

Samajhna zaroori hai, kyunki dono bugs inke beech ke handoff me hain.



Normal 1:1 call flow:

```
 1. Caller taps call
 2. CLIENT  → chat-backend : call:ring            (app plane)
 3. chat-backend → callee's socket : incoming     (agar app live hai)
 4. chat-backend → APNs/FCM push                  (agar app killed/background
 5. CLIENT  → mediasoup : callUser                (media plane)
 6. Caller PRE-JOINS the room while it rings:
        joinRoom → createWebRtcTransport(send/recv) → produce(mic)
        ⚠️  mic is produced PAUSED  (privacy: ring ke dauraan awaaz leak na ho
 7. Callee's device confirms ring  → call:ringing → device_ringing trace
 8. Callee accepts → acceptCall
 9. mediasoup → BOTH sides : callAccepted
10. Caller's SDK: callAccepted handler me producer.resume()   ← 🔴 BUG #1 yahi
11. Callee joins room, consumes caller's producer, resumes consumer
```

```
12. Caller consumes callee's producer, resumes consumer
```

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3. BUG #1 — One-way audio

3.1 Mechanism

Step 6 me caller ka mic **PAUSED** produce hota hai. mediasoup me ek paused producer ka RTP **drop** ho jaata hai — consumer resumed ho to bhi silence forward hoti hai.

Step 10 wala `callAccepted` handler **duniya ka ekmatra** code hai jo us producer ko resume karta hai. Reference (web): `talkstry-web/src/calls/sdk/CallingSDK.js:594`

```
['mic', 'camera'].forEach(function (k) {
  var pr = self._producers[k];
  if (pr && !pr.closed) {
    pr.resume();
    self._syncProducerState(pr);
  }
});
```

Agar ye ek event **lost** ho jaaye, ya apne `mine` / roomId guard (`CallingSDK.js:602`) pe **drop** ho jaaye:

- Caller ka producer poori call paused rehta hai
- **Callee ko caller ki awaaz nahi aati**
- **Caller ko callee ki awaaz aati hai** (callee ka producer kabhi paused nahi hota)
- UI "connected" dikhata hai, duration timer chalta hai, **koi watchdog fire nahi hota**

3.2 Kuch bhi self-heal nahi karta

`_reassertMediaState()` (`CallingSDK.js:241`) hi woh function hai jo producer state dobara assert karta hai. Lekin woh **sirf** do jagah se chalta hai:

1. `_emitUp()` — ICE-down recovery ke baad
2. Socket reconnect (`_resume()`)

Healthy call me in dono me se kuch nahi hota. `_sampleStats()` (`CallingSDK.js:1291`) sirf RTT/jitter/loss collect karta hai — local mute state ko server ke producer state se **compare nahi karta**. Isliye ek baar miss hua to poori call silent.

3.3 Evidence

Broken call `room_call_1792826141359_795` — `[produce] resumed` line kabhi nahi aati:

```
07:15:42.537 [produce] peer=jarvis austrum kind=audio producerId=cb01bfb0…
07:15:42.791 [produce] paused  peer=jarvis austrum            ← pre-answer hol
07:15:54.159 [lobby] acceptCall … by 6a77578326…
07:15:54.589 [consume] peer=soib peter producerId=cb01bfb0…   ← callee pulls i
07:15:54.985 [consume] resumed peer=soib peter                ← consumer theek
             ✗ NO "[produce] resumed peer=jarvis austrum"     ← mic paused for
07:15:55.527 [consume] resumed peer=jarvis austrum            ← caller sunta h
07:16:08.791 [lobby] endCall …                                ← 14s me hangup
```

Working call me yahi line accept ke **73ms** baad aati hai.

Metric	Value
Accepted calls with a pre-answer paused mic	509
Promote resume OK	457
Never resumed (BUG)	**52 (10.2%)**
Inme se call initiator ka mic	**49 of 52**

Broken calls ki duration 4–30 second — logs me caller khud `cancelCall` karta hai, kyunki saamne wala kehta hai "awaaz nahi aa rahi". Kai users ne lagatar 3–4 baar retry kiya (`peter → Govind barshaiya` , 07:45–07:47 me 4 baar).

Server side pe koi anomaly nahi — `emitTo` caller ke saare devices ko `callAccepted` bhejta hai (`lobby.js:366`), aur broken vs working calls me koi server-side difference nahi mila. (`re-delivered ring` correlation check kiya — broken me 96%, par working me bhi 91%, to woh cause nahi hai.) Loss **client pe** hai.

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4. BUG #2 — Calls apne aap cancel ho jaati hain

4.1 Root cause: APNs VoIP push ka IPv6 leg toota hua hai

`talsksty-baackend/logs/calls/` se:

```
APNs attempts (iOS VoIP push)
  primary leg   total=278   ok= 78   FAIL=200    ← 72% fail
                FAIL latency: median 2077ms (min 572, max 2097)
                ok   latency: median  252ms
  ipv4 retry    total=200   ok=200   FAIL=  0    ← 100% success
                ok   latency: median  261ms
```

`~2077ms` ka consistency batata hai ki ye **timeout** hai, random failure nahi — IPv6 address pe connect karne ki koshish, ~2s pe timeout, phir IPv4 fallback jo hamesha kaam karta hai.

Culprit: `talsksty-baackend/src/services/apnsVoipPush.service.js:351`

```
let res = await tryHost(primary, { autoSelectFamily: true }, 'primary');
```

Node 20 ka happy-eyeballs (IPv6+IPv4 raced) is server pe kaam nahi kar raha. IPv4 retry line 365-366 pe hai aur **hamesha** succeed karta hai.

4.2 Pura causal chain

```
1. iOS callee live socket pe nahi → VoIP push chahiye
2. apnsVoipPush.service.js:351 → autoSelectFamily:true
   → ~2.08s baad "The pending stream has been canceled"
3. IPv4 retry → 261ms me success
4. Total push delivery = 2.08s + 0.26s = ~2.34 second
5. Callee ka phone ~2.3–3s pe bajta hai (~0.3s ki jagah)
6. Caller ki app ~2.3s pe dial abort kar deti hai
7. → cancel src=app code=null → CallLog me "cancelled"
8. → 232 of 649 outgoing calls (35.7%)
```

4.3 Supporting evidence

Metric	Value
`[CALL][socket] ring` (total dials)	410
`device ring confirmed`	373
Calls jisme callee ka phone kabhi baja hi nahi	**37 (9%)**
ring → device_ringing latency	median 0s, p75 1s, **p90 3s**, max 18s
2 second ya zyada lagne wali calls	**91 of 373 (24%)** ← app yahin abort karti hai
Ring confirm source	socket 218, **voip 142**, fcm push 30
`ring timeout → no_answer`	**sirf 6** (410 rings me se)

142 ring-confirmations VoIP push se aaye — wahi APNs par depend karte hain.

4.4 Kaun cancel kar raha hai — user ya app?

`calltraces` me `end.source` field se, 250 cancel-ended calls:

source	n	%
`user` — asli user cancel	**66**	26%
`app` / `app_auto` — **automatic**	**184**	**74%**

Sabse bada bucket: **165 calls with `code: null`, `trigger: null`, `message: null`** — app ne call end ki aur koi reason record nahi kiya.

Proof ki ye purana build nahi, app ka alag code path hai:

	n
Devices jo causeless cancel bhejte hain	5
Devices jo proper cause bhejte hain	15
Devices jo DONO karte hain	**2**
Callers jo dono groups me hain	**8 of 22**

Ek hi device kabhi `code:'user_cancel'` bhejta hai aur kabhi `code:null`. Agar build purana hota to woh device **hamesha** causeless hota. Matlab same build me **do alag code paths** hain — ek user-tap ka (cause ke saath), ek app ke automatic abort ka (bina cause).

> ⚠️ Timing distribution se ye prove **nahi** hota — app-cancels aur user-cancels ka timing lagbhag same hai (8s ke andar: 55% vs 53%). Asli proof yahi device-overlap hai.

4.5 Secondary: har unanswered call "cancelled" dikhti hai

Do ring timeouts hain aur client wala pehle fire hota hai:

	Value	File
Client ring timer	~30–35s	`talkstry-web/src/calls/constants.js:80`
Server authoritative `no_answer`	40s + 2s grace = **42s**	`call.handler.js:44,47`

Client server se 7–12s **pehle** give up karta hai, to server ka `no_answer` path (`call.handler.js:690`) kabhi chalta hi nahi — **410 rings me sirf 6 baar**. Upar se

`callManager.js:993` server ke authoritative `no_answer` ko bhi outcome `'cancelled'` bana deta hai.

Result: `no_answer`, `busy`, `unreachable` — teeno enum me maujood hain par `calllogs` me **ZERO rows**. Sab "cancelled" me gir jaata hai, jo user ko "app ne khud cancel kar di" jaisa dikhta hai.

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5. Directory-wise change plan

📁 5.1 `talsksty-baackend` — P0, asli fix yahin hai

P0 — APNs IPv4 first (ye ek change 36% cancel rate theek karega)

File: `src/services/apnsVoipPush.service.js`

Line 351 — `autoSelectFamily: true` ko IPv4 se replace karo:

```
// BEFORE
let res = await tryHost(primary, { autoSelectFamily: true }, 'primary');

// AFTER — IPv4 pehla attempt (data: 200/200 success, median 261ms)
let res = await tryHost(primary, { family: 4 }, 'primary');
```

Aur line ~359-366 ka fallback ulta kar do: agar IPv4 fail ho to `autoSelectFamily: true` try karo (ulta nahi).

Ya global fix — `src/server.js` ke sabse top pe:

```
require('dns').setDefaultResultOrder('ipv4first');
```

> Connection cache key (line 190) me family already included hai, to pool sahi reuse hoga — extra change nahi chahiye.

P1 — Observability + labelling

File	Line	Change
`src/socket/handlers/call.handler.js`	**44**	`RING_WINDOW_SEC` default `40` ko `src/services/call/callAnswerLoc` ke `35` se align karo. Ek hi env var `CALL_RING_DURATION_SECONDS` ke d hain

File	Line	Change
`src/socket/handlers/call.handler.js`	**373**	`source: cr.source || 'app'` → ca hone pe alag marker (`'app_nocaus` `logger.warn` . Isse 165 causeless c diagnosable ho jaayenge
`src/services/call/callLog.service.js`	—	Server ka authoritative `no_answer` baad me aane wale `cancelled` se **se roko**. Isi wajah se DB me `no_an` hain

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📱 5.2 Mobile app — Android / iOS CallingSDK (P0/P1 — aapke paas code hai)

Native SDK is repo me nahi hai, par web SDK wahi architecture mirror karta hai. Neeche **behaviour contract** diya hai + apne code me kahan dhoondhna hai.

M-1 (P0) — Caller ka promote state-convergent banao → BUG #1 fix

Dhoondho: `callAccepted` / `onCallAccepted` handler, aur jahan `preAnswer` / `pendingAnswer` flag clear hota hai.

Problem: aaj resume sirf ek baar, sirf `callAccepted` aane pe hota hai. Woh event miss hua to mic hamesha paused.

Karna kya hai:

1. Ek function banao — `reassertMediaState()` (web reference: `CallingSDK.js:241`) jo:
   - har local producer pe **local truth** (`producer.paused`) dobara server pe assert kare — `pauseProducer` / `resumeProducer`
   - har consumer pe `resumeConsumer` dobara bheje
   - **sirf** `resume` retry mat karo — hamesha `pr.paused` **ab** padho aur wahi assert karo, warna ek late `pause` retry promote ke baad land kar ke call hamesha ke liye mute kar dega
2. Use **in sab jagah** se call karo (aaj sirf ICE-recovery aur socket-reconnect pe hota hai):
   - ✅ `callAccepted` pe (jaisa aaj hai)
   - 🆕 **recv transport** `connected` **hone pe**
   - 🆕 **pehla** `newProducer` **event aane pe** (matlab remote peer room me aa gaya)

- 🆕 Call ACTIVE hone ke **2s baad ek safety tick**
- ✅ socket reconnect pe
- ✅ ICE recovery pe

3. **Guard hatao / dheela karo:** aaj `callAccepted` ek `mine` check pe drop ho jaata hai (web: `CallingSDK.js:602` — `_outIds[cid] || _out.callId === cid || _acceptedId === cid`). Agar dial-id bookkeeping clear ho gayi (grace sweep, redial, reassert) to accept silently drop ho jaata hai. Agar hum **pehle se us roomId me preAnswer** state me baithe hain, to callAccepted ko **roomId** se match karo, callId se nahi.

Acceptance: call server log me har accept ke baad caller ka `[produce] resumed peer=<name>` line **zaroor** aani chahiye.

M-2 (P0) — Dial ko 2–3 second me abort karna band karo → BUG #2 ka client half

Dhoondho: outgoing dial ke paas koi short timer — `ringAckTimeout`, `deviceRingingTimeout`, `reachabilityTimeout`, ya `device_ringing` / `call:ringing` ka wait jo abort trigger karta hai.

Problem: APNs fix ke baad bhi, app ka ~2.3s abort kabhi-kabhi jeet jaayega. Aur yahi 90 of 165 auto-cancels (55%) maar raha hai.

Karna kya hai:

- `device_ringing` ke **na aane** ko abort condition **mat** banao. "Calling..." dikhao aur poora ring window do.
- Dial ka terminal verdict **server** se lo — `call:timeout` / `call:state {state:'ended', reason:'no_answer'}`.
- Agar koi local safety timer rakhna hi hai, to **minimum 10 second**, 2–3s nahi.

Acceptance: `calltraces` me `end.source:'app'` wali cancels me ring duration **< 8s** wali entries khatam ho jaayein.

M-3 (P1) — Har cancel ke saath `cause` bhejo

Dhoondho: jahan se `call:cancel` / `cancelCall` emit hota hai. Ek path user-tap ka hai (cause bhejta hai), doosra automatic hai (cause nahi bhejta).

Karna kya hai: automatic path me bhi cause attach karo:

```
{ "code": "ring_ack_timeout", "trigger": "system", "source": "app_auto" }
{ "code": "device_unreachable", "trigger": "system", "source": "app_auto" }
```

Backend already support karta hai (`call.handler.js:373` — `cr.source || 'app'`).

Acceptance: `calltraces` me `end.source:'app'` + `end.code:null` wali rows **zero** ho jaayein.

M-4 (P1) — Client ka ring timer server se bada rakho

Server authoritative timeout **42s** hai (40 + 2s grace). Client ka timer **45s** karo, ya client se call-end hatao hi aur server ka verdict render karo.

M-5 (P2) — Ring-timeout ko `cancelled` label mat do

Jab ring-out ho (nobody answered), reason `no_answer` bhejo, `cancelled` nahi. (Web me yahi bug `callManager.js:993` aur `:1460` pe hai.)

📁 5.3 `talkstry-web` — P2 (parity ke liye; production impact ~0)

> Web pe production me sirf **10 calls** hain (649 me se). Ye cancel rate nahi girayega — par ye reference implementation hai jise native SDK mirror karta hai, to patch yahan likho aur wahan le jaao.

File	Line	Change
`src/calls/sdk/CallingSDK.js`	**594**, **241**	**BUG #1 fix** — M-1 jaisa: `_reassertMediaState()` ko recv-transport connect + pehle `newProducer` pe bhi chalao
`src/calls/constants.js`	**80**	`DEFAULT_RING_SECONDS = 35` → **45** (server ke 42s se bada)
`src/calls/callManager.js`	**1460**	ring timer khud call end karna band kare; server ka `call:timeout` wait kare
`src/calls/callManager.js`	**993**	server ke `no_answer` ko `'cancelled'` pe map karna band karo
`src/calls/callManager.js`	**1349**	`cancelCall({callId, toUserIds})` me `cause` add karo (abhi bilkul missing)

🖥️ 5.4 Call server `/var/www/mediasoup` — P3 (observability)

Server: `ssh -p 2259 root@151.185.45.135`

File	Line	Change
`src/lobby.js`	**260-266**	`user offline / X is busy / you are already in a call` — teeno refusals pe `console.log` add karo. Abhi ye dials logs me **completely invisible** hain
`src/lobby.js`	**749**	Grace-sweep ka silent `callEnded` emit — log line add karo. Ye caller ko bina kisi trace ke bhejta hai

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6. Kaise verify karein

6.1 BUG #1 — One-way audio

Call server pe. Jis call me `paused` ke baad `resumed` **nahi** hai — wahi one-way call hai:

```
ssh -p 2259 root@151.185.45.135 \
  'grep -E "\[produce\] (paused|resumed)|acceptCall" \
   /root/.pm2/logs/calling-backend-out-0.log | tail -60'
```

Poora scan (kitni calls broken hain):

```
# server pe ye script chalao — "never resumed" count deta hai
ssh -p 2259 root@151.185.45.135 'python3 - <<"PY"
import re,collections
from datetime import datetime
LOG="/root/.pm2/logs/calling-backend-out-0.log"
def ts(s):
    try: return datetime.strptime(s,"%Y-%m-%dT%H:%M:%S.%fZ")
    except: return None
rp=re.compile(r"\[produce\] room=(\S+) peer=(.*?) kind=audio source=mic produc
rq=re.compile(r"\[produce\] (paused|resumed) peer=(.*?) kind=audio producerId=
ra=re.compile(r"\[lobby\] acceptCall (\S+) by (\S+) \(room (\S+)\)")
rc=re.compile(r"\[consume\] room=(\S+) peer=(.*?) kind=audio producerId=(\S+)'
P={};A={}
for l in open(LOG,errors="replace"):
    t=ts(l.split(" ",1)[0]) if l[:4].isdigit() else None
    if not t: continue
    m=rp.search(l)
    if m: P[m.group(3)]={"t0":t,"room":m.group(1),"peer":m.group(2).strip(),"e
    m=rq.search(l)
    if m and m.group(3) in P: P[m.group(3)]["ev"].append((t,m.group(1))); cont
    m=rc.search(l)
    if m and m.group(3) in P: P[m.group(3)]["c"].append(t); continue
    m=ra.search(l)
    if m: A[m.group(3)]=t
bad=ok=0
```

```
for pid,d in P.items():
    if d["room"] not in A or not d["ev"]: continue
    if d["ev"][0][1]!="paused" or (d["ev"][0][0]-d["t0"]).total_seconds()>3: c
    if not d["c"]: continue
    if any(e[1]=="resumed" for e in d["ev"]): ok+=1
    else: bad+=1; print("BROKEN", d["room"], d["peer"])
print(f"\npromote OK={ok}  NEVER RESUMED={bad}")
```

Fix ke baad expected: `NEVER RESUMED = 0`

6.2 BUG #2 — APNs latency

`talsksty-baackend/logs/calls/` me:

```
cd /var/www/whatsapp/talsksty-baackend/logs/calls

(zcat *.gz; cat call-$(date +%Y-%m-%d).log) 2>/dev/null | python3 -c "
import sys,json,collections,statistics
att=collections.defaultdict(list)
for l in sys.stdin:
    try: d=json.loads(l)
    except: continue
    if str(d.get('message',''))!='[VOIP] APNs attempt': continue
    att[d.get('attempt')].append((d.get('ms'),d.get('status')))
for a,v in att.items():
    ok=[m for m,s in v if s==200]; bad=[m for m,s in v if s!=200]
    print(f'{a:9s} total={len(v):4d} ok={len(ok):4d} fail={len(bad):4d}')
    if ok:  print(f'          ok  median={statistics.median(ok):.0f}ms')
    if bad: print(f'          FAIL median={statistics.median(bad):.0f}ms')
"
```

Fix se pehle: `primary total=278 ok=78 fail=200`, FAIL median ~~2077ms~~
~~Fix ke baad expected:~~ ~~`primary`~~ ~~leg ka fail count **0**~~, median ~260ms, aur `ipv4` retry leg lagbhag khali

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6.3 Ring latency (callee ka phone kab bajta hai)

```
cd /var/www/whatsapp/talsksty-baackend/logs/calls

(zcat *.gz; cat call-$(date +%Y-%m-%d).log) 2>/dev/null | python3 -c "
import sys,json,collections,statistics
from datetime import datetime
def T(s):
    try: return datetime.strptime(s,'%Y-%m-%d %H:%M:%S')
```

```
    except: return None
ring={};conf={}
for l in sys.stdin:
    try: d=json.loads(l)
    except: continue
    m=str(d.get('message','')); t=T(d.get('timestamp','')); c=d.get('callId')
    if m=='[CALL][socket] ring' and c and c not in ring: ring[c]=t
    elif m=='[CALL][ringing] device ring confirmed' and c and c not in conf: c
lat=sorted((conf[c]-ring[c]).total_seconds() for c in ring if c in conf and cc
print('rings',len(ring),' confirmed',len(lat),' NEVER rang',len(ring)-len(lat)
if lat: print('median',statistics.median(lat),'p90',lat[int(len(lat)*.9)],'max
b=collections.Counter()
for x in lat: b['0-1s' if x<1 else '1-2s' if x<2 else '2-3s' if x<3 else '3-5s
print(dict(b))
"
```

Fix se pehle: p90 = 3s, 24% calls ≥ 2s, 37 of 410 never rang
Fix ke baad expected: p90 ≤ 1s, "never rang" count kaafi kam

6.4 Cancel rate (MongoDB)

```
mongosh "mongodb://chat:<PASSWORD>@164.52.219.164:27017/chat?directConnection=
const since = new Date(Date.now() - 7*24*3600*1000);
print("=== outcome (outgoing, last 7d) ===");
printjson(db.calllogs.aggregate([
  {$match:{createdAt:{$gte:since}, direction:"outgoing"}},
  {$group:{_id:"$reason", n:{$sum:1}}},{$sort:{n:-1}}
]).toArray());
print("=== cancels: user vs app ===");
printjson(db.calltraces.aggregate([
  {$match:{createdAt:{$gte:since}, "end.action":"cancelled"}},
  {$group:{_id:{src:"$end.source",code:"$end.code"}, n:{$sum:1}}},{$sort:{n:-1
]).toArray());
'
```

Fix se pehle: `cancelled` 232 / 649 (35.7%); cancels me `source:'app'` + `code:null` = 165
Fix ke baad expected: `cancelled` ~10%, `no_answer` rows dikhne lagein, `code:null` rows = 0

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7. Execution order (checklist)

```
[ ] STEP 1 — P0 backend, 1 line  (36% cancel rate ka asli ilaaj)
    [ ] talsksty-baackend/src/services/apnsVoipPush.service.js:351
```

```
        autoSelectFamily:true  →  family:4
    [ ] deploy + restart
    [ ] §6.2 se verify: primary leg fail count ~0
    [ ] §6.3 se verify: ring latency p90 ≤ 1s
    [ ] 24h baad §6.4 se cancel rate check

[ ] STEP 2 — Observability (koi behaviour change nahi, sirf visibility)
    [ ] talsksty-baackend/src/socket/handlers/call.handler.js:373  (cause-miss
    [ ] talsksty-baackend/src/services/call/callLog.service.js     (no_answer
    [ ] /var/www/mediasoup/src/lobby.js:260-266  (silent refusals log)
    [ ] /var/www/mediasoup/src/lobby.js:749      (silent callEnded log)
    [ ] Mobile: M-3 (har cancel ke saath cause)
    [ ] 2-3 din data collect karo

[ ] STEP 3 — Mobile client fixes
    [ ] M-2  dial ka 2-3s abort hatao           (P0)
    [ ] M-1  promote state-convergent banao     (P0 — BUG #1)
    [ ] M-4  client ring timer 45s              (P1)
    [ ] M-5  ring-out ko no_answer label do     (P1)

[ ] STEP 4 — Timeout alignment
    [ ] talsksty-baackend/src/socket/handlers/call.handler.js:44  ↔  callAnswe
    [ ] talkstry-web/src/calls/constants.js:80       → 45
    [ ] talkstry-web/src/calls/callManager.js:1460, 993, 1349

[ ] STEP 5 — Web parity (optional, traffic ~0)
```

Expected outcome

Metric	Aaj	Fix ke baad
Cancel rate (outgoing)	**35.7%**	~10% (asli user-cancel rate)
One-way audio calls	**10.2%**	~0%
APNs primary leg failures	**72%**	~0%
Ring latency p90	**3s**	≤ 1s
Callee ka phone kabhi na bajna	**9%**	~2%
`no_answer` rows in DB	**0**	properly recorded

---

8. Alag se: S3 recording uploads fail ho rahe hain

Calling se related nahi, par check me mila:

```
[s3] upload failed for rec_1792829378071_51:
     The AWS Access Key Id you provided does not exist in our records.
```

`/var/www/mediasoup/.env` ka `S3_ACCESS_KEY` invalid / rotated hai. Recordings local pe ban rahi hain par S3 pe upload nahi ho rahi (`S3_DELETE_LOCAL=true` hai, to check karo ki files delete na ho rahi hon).

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9. File reference (quick index)

Bug / Fix	Directory	File:Line
APNs IPv6 failure (P0)	`talsksty-baackend`	`src/services/apnsVoipPush.ser`
APNs IPv4 retry (reference)	`talsksty-baackend`	`src/services/apnsVoipPush.ser`
Ring window default mismatch	`talsksty-baackend`	`src/socket/handlers/call.hanc` `src/services/call/callAnswerL`
Cause-missing default	`talsksty-baackend`	`src/socket/handlers/call.hanc`
Server no_answer path	`talsksty-baackend`	`src/socket/handlers/call.hanc`
Push fan-out (already correct)	`talsksty-baackend`	`src/socket/handlers/call.hanc`
Reason → outcome mapping	`talsksty-baackend`	`src/services/call/callReason.`
CallLog schema	`talsksty-baackend`	`src/models/mongo/call/callLog`
One-way audio: promote	`talkstry-web`	`src/calls/sdk/CallingSDK.js:5`
One-way audio: pre-answer pause	`talkstry-web`	`src/calls/sdk/CallingSDK.js:1`
One-way audio: reassert fn	`talkstry-web`	`src/calls/sdk/CallingSDK.js:2`
callAccepted `mine` guard	`talkstry-web`	`src/calls/sdk/CallingSDK.js:6`
Client ring timer	`talkstry-web`	`src/calls/constants.js:80` , `src/calls/callManager.js:1460`
no_answer → cancelled mislabel	`talkstry-web`	`src/calls/callManager.js:993`

Bug / Fix	Directory	File:Line
Causeless cancelCall	`talkstry-web`	`src/calls/callManager.js:1349`
Silent dial refusals	`/var/www/mediasoup`	`src/lobby.js:260-266`
Silent callEnded	`/var/www/mediasoup`	`src/lobby.js:749`
callAccepted fan-out	`/var/www/mediasoup`	`src/lobby.js:366`
SFU produce/consume/resume	`/var/www/mediasoup`	`src/signaling.js`

10. Access details

Resource	Address
Call server (mediasoup)	`ssh -p 2259 root@151.185.45.135 → /var/www/mediasoup`
Call server logs	`/root/.pm2/logs/calling-backend-out-0.log`
chat-backend call logs	`talsksty-baackend/logs/calls/call-YYYY-MM-DD.log[.gz]`
MongoDB	`164.52.219.164:27017/chat` — collections `calllogs`, `calltraces`
Recording webhook	`https://backend.talkstry.com/api/v2/internal/call/recording`

> Credentials `talsksty-baackend/.env` aur `/var/www/whatsapp/.env` me hain — is file me deliberately nahi likhe.